

Episcap: project log

Brandon Neff

John Altin

TGen

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Living record. Read top to bottom to get up to speed. Figures regenerated from `results/`; see `docs/WORKFLOW.md`

This is the running record of the episcaf project: scaffolding antibody epitopes defined by 3D geometry into folded, assay-ready designs (RFDiffusion3 $\rightarrow$ ProteinMPNN $\rightarrow$ AlphaFold3). It is meant to be read in order, since each section is one iteration: what we tried, what we found, and what it changed for the next step. It is not a formal manuscript. Every figure is regenerated by a named script (recorded in `manuscript/figures/FIGURES.md`) and every number traces to `results/`; unconfirmed values are flagged [verify].

1 Motivation: scaffolding B-cell epitopes

1.1 Why scaffold epitopes

This project is about *B-cell epitopes*: the regions of an antigen that an antibody recognizes. Antibodies recognize these regions through their three-dimensional geometry, and many of the most interesting B-cell epitopes are discontinuous in sequence but contiguous in space, meaning the contacted residues are scattered along the antigen sequence yet fold together into a single surface patch. This is precisely the kind of target that linear methods cannot represent. High-throughput serology platforms such as PepSeq tile an antigen into short, overlapping linear peptides, which works well for continuous, sequence-local epitopes. A linear peptide, however, cannot reconstitute a surface assembled from sequence-distant residues, and it discards the native conformation the antibody evolved to recognize.

The central question of this project is whether *scaffolding* an epitope produces a stronger, cleaner binding signal in the PepSeq assay than linear tiling does. By embedding the epitope residues in a folded protein, we can hold them in the conformation the antibody actually sees, and we can present distinct epitopes as separable, individually testable designs. If that conformational presentation matters, scaffolded designs should bind their cognate antibodies more reliably than the corresponding linear peptides. The longer-term goal is to extend PepSeq-style high-throughput assays beyond linear peptides to conformational, discontinuous epitopes.

1.2 The approach

We extract the epitope, meaning the residues an antibody contacts, from an antibody:antigen crystal structure. We embed those residues into a *de novo* protein scaffold with RFDiffusion, assign a sequence with ProteinMPNN, and then ask AlphaFold3 whether the design actually folds with the epitope held in its native geometry. Designs are kept or rejected on AlphaFold3 confidence and geometry-preservation metrics (Section 2). The full production pipeline, including the Nextflow

orchestration, the RFdiffusion runs, and the AlphaFold3 evaluation, originates in Lawson Woods’ `episcaf-experiments` repository. Reproducing and then extending it is the starting point of this work.

1.3 Patches and islands

A single antibody:antigen complex can yield more than one epitope *patch*, labeled 0P, 1P, and so on, where each patch is a distinct cluster of antibody-contacting residues drawn from the same structure. A patch is in turn made of one or more *islands*, the spatially contiguous runs of epitope residues in three-dimensional space. This distinction matters because it predicts difficulty. Single-island patches can be enforced locally and tend to scaffold well, whereas multi-island patches require preserving the relative position and orientation of spatially separated regions, which is much harder. A concrete failure makes the point: PDB 2XQB, patch 0P, is a 56-residue epitope split across two separated islands, and across 2,624 RFdiffusion trials zero designs passed the downstream filters. Because this 0% rate persists over thousands of attempts, it reads as a robust computational failure rather than undersampling. The cause appears procedural: contigs (the constraints RFdiffusion is given) enforce epitope residue identity and ordering, but they do not strongly constrain the distances and orientations *between* islands, so the model can satisfy the contig and still lose the geometry an antibody needs.

1.4 Two design pools: DP3 and DP4

The project runs in two settings that differ in how much we know about the antibody.

- **DP3 (known antibody).** When the antigen has a solved antibody complex, we know the exact epitope and we have a ground truth for “good”: the design must hold the epitope in place and must not place scaffold mass where the antibody has to bind. This is the setting where we have already run designs through the assay and seen some success, and it is where the question of scaffolded-versus-linear signal is most directly testable.
- **DP4 (tiled 12-mers, no known antibody).** For most antigens we have no solved antibody, so we do not know where the epitope is. Here we instead tile the antigen sequence into overlapping 12-residue windows, stepping by two residues at a time, and scaffold each window. The aim is to cover the antigen densely enough to catch whatever the real epitopes turn out to be.

1.5 Why a cylinder: the antibody-clash filter without an antibody

One of the strictest filters in the DP3 setting concerns antibody accessibility. We superimpose the design’s epitope region onto the native antigen, place the real antibody where it sits in the crystal complex, and count scaffold residues that clash into the antibody. Any design with a clash count above zero is rejected, because scaffold mass in the antibody’s path would block the very binding event we are trying to recreate. This filter is only available for DP3, where a real antibody exists. For DP4 there is no antibody to clash against, so we cannot apply it directly. The native-antigen-aware cylinder surrogate (Section 6) is our attempt to recover this filter without an antibody: it approximates the volume an approaching antibody would occupy and asks whether scaffold mass intrudes into it. A single, config-driven composite scorer ranks designs in both pools, differing only in which accessibility term it uses.

1.6 How the rest of this log reads

The sections that follow are in the order the work happened. We start by reproducing Lawson’s workflow and confirming that our reimplementations of the metrics matches his, then compare the RFDiffusion1 and RFDiffusion3 design sets directly (Section 2, Section 3). We then turn to the filtering, where the original global metrics bias selection toward rigid, well-ordered folds, and show how decomposing them widens the designable space (Section 4). From there we build a composite score that selects the best designs per epitope (Section 5), and finally extend everything to DP4 using the native-aware cylinder surrogate (Section 6). Each step motivates the next.

2 The pipeline, and reproducing Lawson’s workflow

2.1 Pipeline

RFDiffusion3 generates scaffold backbones around the epitope, ProteinMPNN re-sequences them, and AlphaFold3 predicts the structure single-sequence (`--norun_data_pipeline`, one diffusion sample, seed 1). Per-design metrics are computed from the AlphaFold3 prediction against the designed structure. Generation code lives in `episcaf_pipeline/`, and the metrics, scoring, and visualization code in `episcaf_analysis/`.

2.2 Reproducing Lawson’s baseline

Before trusting any new number we re-implement Lawson’s metrics and check them against his ground truth. `compute_metrics.py --validate` reads `dp2.parquet` (Lawson’s stored per-design values), recomputes `overall_rmsd` and `epitope_chunk_rmsd` from his ProteinMPNN PDBs and AlphaFold3 CIFs, and diffs them against the stored values. Agreement on this diff is the precondition for everything downstream: it confirms that our implementation of the filters matches the workflow we are extending, so that later differences reflect the design protocol rather than a scoring bug.

2.3 Four-filter ground truth (DP3 set)

A design passes when all four conditions hold: overall backbone RMSD ≤ 2 Å, epitope-chunk RMSD ≤ 1 Å, mean predicted aligned error < 5 , and zero antibody residues within 4 Å of non-epitope scaffold residues (`af3_n_clash_res`). Clash detection Kabsch-aligns the AlphaFold3 epitope onto the true epitope, then checks non-epitope scaffold heavy atoms against antibody atoms. The antibody coordinates always come from the AbDb crystal structure, not from AlphaFold3.

2.4 Decomposed metrics

The two continuous filters, overall RMSD and mean PAE, treat the epitope and the scaffold as a single unit. We split each into epitope- and scaffold-specific terms (epitope RMSD, scaffold RMSD, epitope PAE, scaffold PAE), retaining the zero-clash hard filter throughout. Decomposition lets selection reward a well-placed epitope on a flexible scaffold (or the reverse) instead of demanding global rigidity, and it lets us ask whether the optimal rigidity profile is epitope-dependent rather than universal (Section 4). Selection over the four metrics is organized as a 2×2 of strict and relaxed thresholds on (epitope, scaffold), with a flexible/flexible arm as a negative control.

2.5 Native-antigen-aware cylinder surrogate (12-mer set)

With no antibody available, we approximate the occluded volume with a cylinder fitted to the epitope. The epitope C α atoms define a plane by SVD, and the outward normal (the antibody approach direction) orients a cylinder of radius 16 Å, height 40 Å, and offset -4 Å. We count the non-epitope scaffold C α atoms inside the cylinder. The native-aware variant excludes scaffold atoms within a short distance of a native-antigen heavy atom, i.e. residues sitting where the native antigen already sits, which a real antibody tolerates. The geometry core is `episcaf_analysis/native_cylinder_core.py` (self-tested).

2.6 Composite scorer

One config-driven scorer (`episcaf_analysis/score.py`) ranks designs in four steps: (i) **gate** on epitope-chunk RMSD; (ii) transform each metric to a higher-is-better score within a scope (pooled, or per antigen); (iii) **weighted sum**; and (iv) keep the **top- k per epitope**. Each metric is one input with its own activation (percentile, min-max, z-score, or sigmoid), so the model is effectively a single-layer perceptron with hand-set weights, feed-forward and without backpropagation. The dials live in `episcaf_analysis/presets.py`:

metric	weight
epitope-chunk RMSD	0.35
epitope PAE	0.25
accessibility	0.25
overall RMSD	0.15

The weights sum to one (renormalized over whichever metrics a given table actually contains). The only term that changes between the two regimes is accessibility: it is `cylinder_native_aware` for the no-antibody 12-mer set and `cylinder_ca_clashes` for the DP3 set, using the same cylinder geometry, with the native-aware carve applied only where there is no real antibody to clash against. The 12-mer preset transforms metrics per antigen and keeps the top 5 designs per epitope, while the DP3 preset transforms pooled and keeps the top 15. The weights are a hand-set starting point; Section 7 describes fitting them to the DP3 `is_pass` label.

3 RFD1 versus RFD3 on the same epitopes

With the metrics reproduced, we compared the two backbone generators head to head: the same DP3 epitopes run through RFDiffusion1 (Lawson’s set) and RFDiffusion3, each followed by ProteinMPNN and AlphaFold3, and scored against the identical four filters. The RFD1 data here is Lawson’s deposited set (`dp2.parquet`), which is missing AlphaFold3 results for 26 870 of its 151 232 designs; that is a property of his dataset rather than of our pipeline, so we compare over designs that actually have an AF3 result. On that basis the two are comparable, with RFD1 modestly higher: RFD1+MPNN passes 840/124 362 (0.68%) and RFD3+MPNN 751/150 936 (0.50%). RFD3 is therefore at least an even trade for RFD1 on this task, which is the basis for standardizing on RFD3 going forward. The per-metric distributions are broadly comparable (Fig. 1), with RFD3 placing somewhat more mass at low epitope and overall RMSD.

The aggregate pass counts hide a more interesting difference in *where* the passes land (Fig. 2). RFD1’s passes are highly concentrated: a single target (`7ox3_0P`) accounts for 630 of its 840 passes, and just two targets account for the large majority. RFD3 spreads its 751 passes across more targets

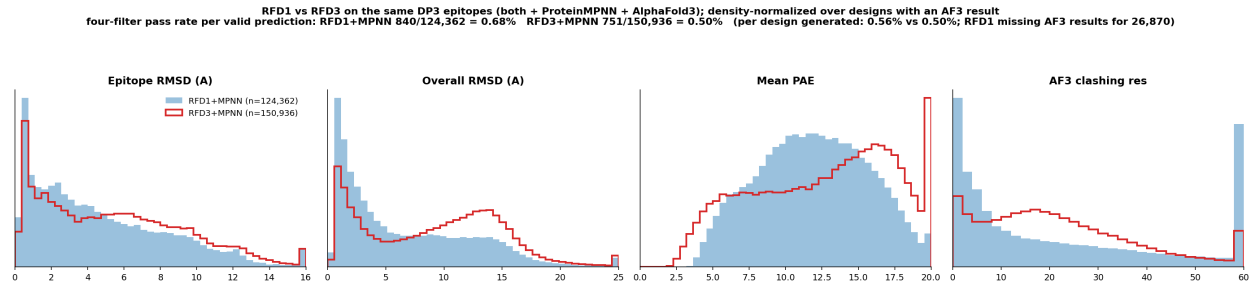


Figure 1: RFD1+MPNN (blue, filled) versus RFD3+MPNN (red, outline) on the same DP3 epitopes, across epitope RMSD, overall RMSD, mean PAE, and AF3 clashing residues, density-normalized over designs with an AF3 result (per-trial n in the legend). Pass rates and the missing-prediction asymmetry are noted in the title. Regenerated by `episcaf_analysis/viz/plot_rfd1_vs_rfd3.py`.

(16 with at least one pass, versus 10 for RFD1), and its per-target leaders are different (4xwo_5P, 8pww_OP, and 8db4_OP all pass well under RFD3 but barely or not at all under RFD1). For a library meant to cover many epitopes, that broader coverage is arguably more useful than a similar total concentrated on one target.

4 Rethinking the filters: global metrics bias toward rigid folds

With the pipeline reproduced and trusted (Section 2), the first thing that needed rethinking was the filtering. The original filters score a design with two global continuous metrics, overall RMSD and mean PAE, that treat the epitope and the scaffold as one unit. This is a reasonable inheritance from standard de novo practice rather than an arbitrary choice, but it biases selection toward globally rigid, well-ordered folds and silently rejects designs whose scaffold is flexible while the epitope is well positioned (or the reverse). Decomposing each metric into epitope- and scaffold-specific terms (Section 2.4) is what lets us see, and eventually recover, that rejected-but-viable space.

To compare design sets on a common footing we use epitope-chunk RMSD < 2.5 Å as a single bar. All three tiled 12-mer antigens pass it more often than the pooled DP3 set:

set	pass rate (< 2.5 Å)	median epitope RMSD (Å)
DP3 / mAb (pooled)	30.0%	5.00
6M0J	47.0%	2.65
1D2K	60.2%	1.85
4WAT	84.4%	0.43

Even the hardest 12-mer antigen (6M0J) clears the bar more often than DP3. The full six-metric distribution comparison is Fig. 3.

Caveats (important for interpretation). This comparison is not yet apples-to-apples. The 12-mer epitope is a single contiguous 12-residue stretch, whereas DP3 epitopes are larger and often multi-island, so 12-mer RMSD/PAE are mechanically lower in part by definition, and the pooled DP3 30% also averages over heterogeneous epitopes. The clean control is to stratify within DP3 by island count and size (Section 7). Cylinder counts are likewise not comparable across sets as absolute numbers. The 12-mer cylinder medians (~ 31 – 32) exceed DP3 (~ 21) largely because the

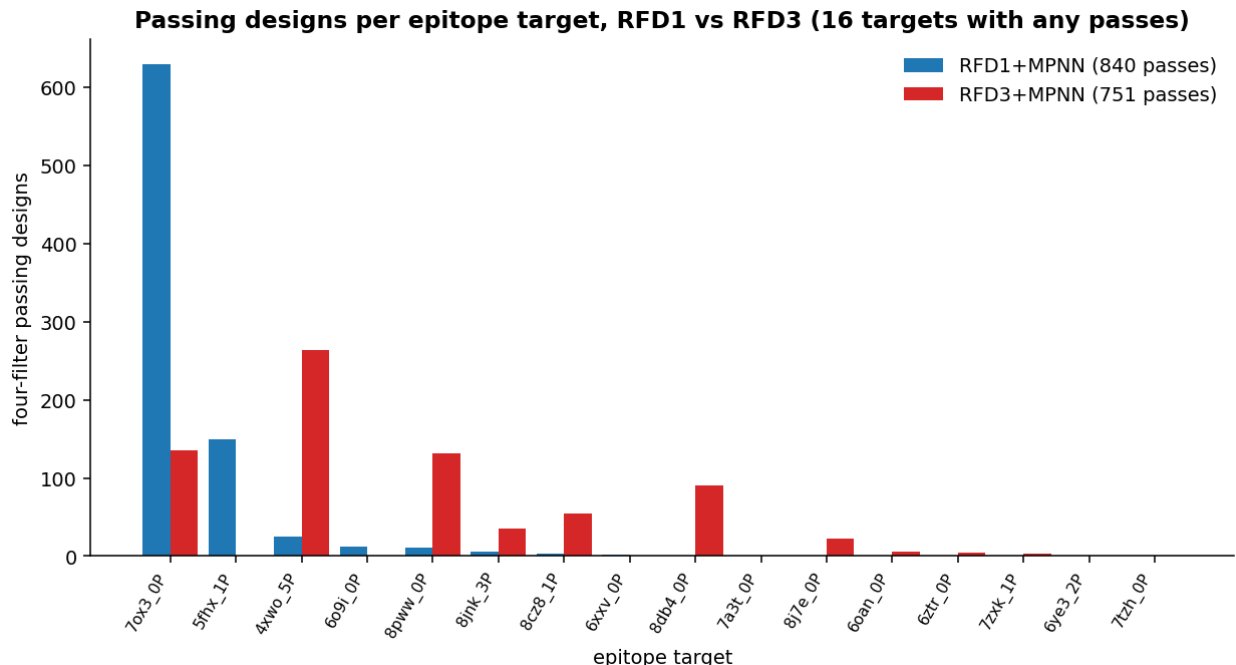


Figure 2: Four-filter passing designs per epitope target, RFD1+MPNN (blue) versus RFD3+MPNN (red). RFD1 is dominated by 7ox3_0p; RFD3 distributes passes across more targets. Regenerated by `episcaf_analysis/viz/plot_passes_per_epitope.py`.

designs are a fixed length (~ 104 residues) while the 12-mer epitope is small (12 residues): that leaves ~ 91 non-epitope scaffold residues whose $C\alpha$ atoms can fall in the fixed cylinder, versus fewer for DP3’s larger epitopes. It is a scaffold-mass artifact rather than more genuine clashing, which is why the score applies the native-aware carve and ranks the cylinder per antigen rather than against an absolute cutoff. As a sanity check, the DP3 cylinder median (≈ 21) tracks its real `af3_n_clash_res` median (≈ 19).

Global versus epitope PAE. The PAE column is a clean illustration of why decomposition matters, and the two are easy to confuse. The four-filter (and Lawson’s original) uses the *global* mean PAE (< 5); on this DP3 set that distribution sits high, median 13.2 with only $\sim 7\%$ of designs below 5. That global value is dragged up by the *scaffold* region, which AlphaFold3 predicts less confidently. The epitope region itself is much better resolved (epitope PAE median 9.1, $\sim 23\%$ below 5). A design can therefore hold its epitope well and still fail the global PAE filter on scaffold uncertainty that may not matter for binding, which is precisely the bias the decomposed metrics are meant to relax. For this reason the figure shows both columns, and the composite scores the epitope-specific PAE rather than the global one.

5 Composite scoring and per-epitope selection

Decomposed metrics give more knobs, which raised the next question: how to combine them into a single ranking so we can take the best designs per epitope. The answer is the config-driven composite scorer (Section 2). It gates on epitope-chunk RMSD, transforms each metric to a higher-is-better score within its scope (per antigen for the 12-mer set, pooled for DP3), weight-sums, and

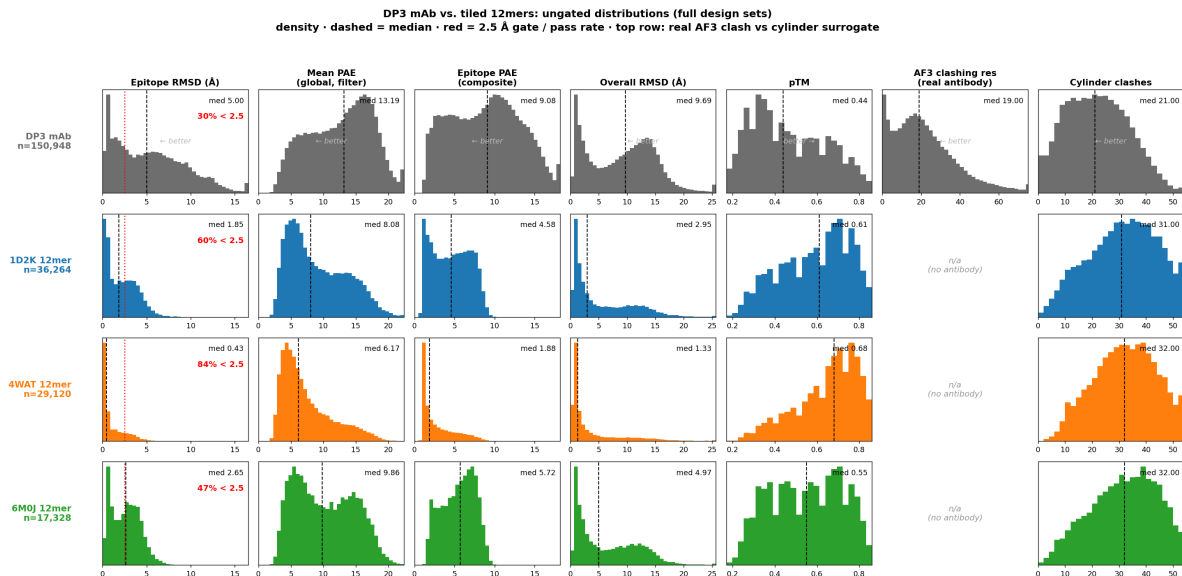


Figure 3: Ungated metric distributions (full design sets), DP3 mAb versus the three 12-mer antigens, across epitope RMSD, global mean PAE (the four-filter metric), epitope PAE (the decomposed metric the composite scores on), overall RMSD, pTM, AF3 clashing residues (antibody set only), and cylinder clashes. Dashed line marks the median; red marks the 2.5 Å epitope-RMSD gate with the per-set pass rate. Regenerated by `notebooks/dp3_vs_12mer_distributions.ipynb`.

keeps the top designs per epitope. Run over the full 12-mer set, the gate keeps 54 539 of 82 712 designs ($\approx 65.9\%$ pooled) before ranking [regen], consistent with the per-antigen rates above.

The scorer is deliberately simple: one metric per input, each with its own activation, weighted and summed, which makes it a single-layer perceptron with hand-set weights. It is still in flux. The per-metric transform is currently a percentile within the population, so a design’s score depends on the cohort it is ranked against. Whether to switch to a population-independent sigmoid (an absolute, saturating per-metric score) is an open choice carried in Section 7.

6 Scaffolding without an antibody: the cylinder surrogate

Most targets of interest have no solved antibody, so the four-filter ground truth does not apply, because there is no antibody to clash against. We extend the workflow to these antigens by tiling them into 12-residue epitopes and replacing the antibody-clash filter with a native-antigen-aware cylinder surrogate (Section 2): a cylinder fitted to the epitope plane along the antibody-approach normal that counts non-epitope scaffold mass intruding into the volume an approaching antibody would need (Fig. 4).

A naive cylinder over-penalizes, because some scaffold mass sits exactly where the native antigen itself already sits, a volume a real antibody clearly tolerates since it binds the native antigen there. The native-aware curve fixes this: among truly clash-free designs, we exclude scaffold C α atoms within 1 Å of a native-antigen heavy atom. Concretely, of the 3186 DP3 designs with zero true antibody clash, the number the plain cylinder flags as heavily penalized (count ≥ 10) falls from 1112 to 384 once the native footprint is carved out, which is 728 **false-positive penalties removed**, while genuinely clashing designs are left essentially untouched (Figs. 5 and 6).

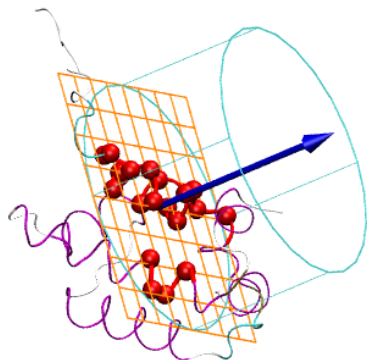


Figure 4: The accessibility cylinder. Epitope residues (red) sit on the scaffold (magenta); the epitope C α atoms define a plane (orange grid) whose outward normal (blue arrow, the antibody-approach direction) orients the cylinder (cyan). Scaffold C α atoms inside the cylinder count against accessibility. Rendered in VMD by `known_antigen/analysis/cylinder_vis/visualize_cylinder.tcl`.

A pipeline constraint worth recording: crystal gaps. Epitope fidelity for the 12-mer set is measured by aligning the AlphaFold3 epitope onto the native crystal epitope, matched by residue number (`build_12mer_metrics.py`). When a tile’s epitope residues are unresolved in the crystal, meaning a gap in the deposited structure, the native C α anchors are missing, the Kabsch alignment cannot be formed, and the tile is dropped with status `crystal_epi_incomplete`. This is a real constraint on which 12-mer epitopes are scorable at all: an antigen whose crystal has gaps over the tiled region contributes fewer (or no) scorable tiles, independent of design quality. The mechanism is confirmed in code; the specific antigen(s) and tile counts affected are not in the local ok-only export and remain to be pulled from the cluster pre-filter status counts [`verify`].

6.1 Interpreting the cylinder: where passing designs land

The cylinder is easiest to read by overlaying, for each metric, the full design population against the designs we would actually advance (Fig. 7). For DP3 the advanced set is the four-filter passes (751 designs); for DP4 it is the top three per epitope by composite score. On every geometry and confidence metric the passing designs concentrate where expected: low RMSD, low PAE, and high pTM. The cylinder column is the informative one. The passing DP3 designs, which we know the antibody accepts, sit at low cylinder counts (median near 10), whereas the DP4 top-3 sit higher (medians near 25–30). This is the same scaffold-mass effect noted earlier rather than a sign that the DP4 designs are worse: with a 12-residue epitope on a $\sim$ 104-residue design, more non-epitope scaffold mass can fall in the fixed cylinder even among the designs we would pick. It is a direct argument for leaning on the native-aware carve and per-antigen scoring rather than an absolute cylinder cutoff shared across sets, and the comparison gives us a concrete reference for what a

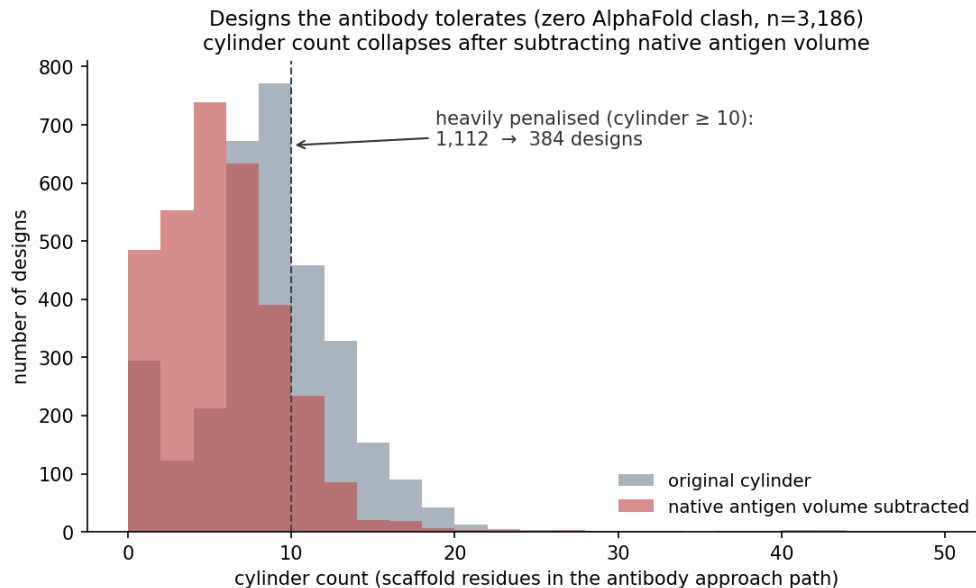


Figure 5: Cylinder count among clash-free DP3 designs (zero AlphaFold3 antibody clash, $n = 3186$), before (grey) and after (red) subtracting the native-antigen volume. The heavily penalized tail (count ≥ 10) collapses from 1112 to 384. Regenerated by `episcaf_analysis/viz/plot_fp_reduction.py --native metrics_native_cyl.csv --mode dist`.

tolerable cylinder count looks like (the DP3 passing range).

The cylinder earns this role. On DP3, where the true antibody clash is known, the cylinder predicts clash-free designs (`af3_n_clash_res=0`) with AUC 0.93 (`cylinder_native_aware` 0.935, slightly better than the plain count 0.924; clash-free vs clashing medians 5 vs 22). It is therefore a strong stand-in for the filter we lose on DP4, and the native-aware carve helps the ranking, not just the optics (`episcaf_analysis/cylinder_surrogate_auc.py`).

7 Open questions and next steps

1. **Within-DP3 island stratification (the clean control).** Split the DP3 set by epitope island count and size and recompute pass rates, to separate the “contiguous-12 is easier” confound from a real scaffolding-difficulty effect. This needs the per-epitope island metadata in `datasets/dp2.parquet`.
2. **Move to a sigmoid transform (planned).** The composite currently maps each metric to a percentile within its population, which is non-differentiable and population-dependent, so it cannot be fit and a score does not transfer from one set to another. A sigmoid (fixed midpoint and slope k per metric) gives an absolute, saturating, differentiable score, which both lets the weights be learned and lets a DP3-trained score transfer to DP4. The scorer already supports it (`transform=sigmoid` in `presets.py`); midpoints can be seeded from the DP3 separations (e.g. a cylinder midpoint near 15, between the clash-free and clashing medians). This is the prerequisite for fitting; the percentile version stays the default until then.
3. **Defer fitting until we have an experimental label.** The composite is a single-layer

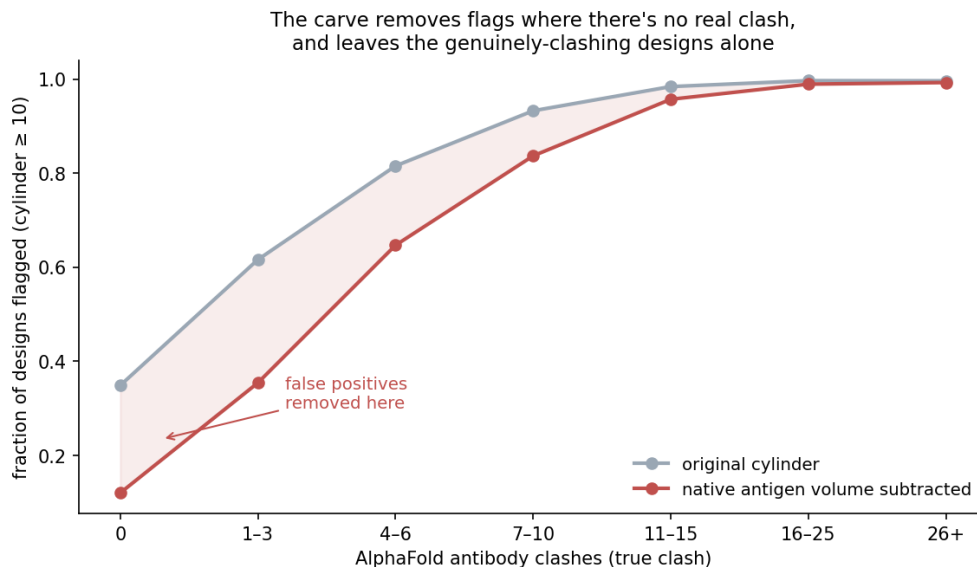


Figure 6: The carve removes flags selectively. Fraction of designs flagged (cylinder ≥ 10) across bins of true antibody clash, before versus after the carve. The gap is large where there is no real clash (false positives removed) and closes where the clash is real (true signal retained). `--mode selectivity`.

perceptron with hand-set weights (feed-forward, no backprop), and the natural next step is to learn the weights. We are deliberately *not* fitting yet: the only label available now is the in-silico `is_pass`, which is defined by the four filters themselves, so fitting to it would largely re-derive the filters. The plan instead is to generate designs at scale, prioritize them with the current composite, run the top picks through the PepSeq assay, and fit the weights against *experimental binding*, the non-circular target. The cylinder weight in particular is already justified by data: on DP3 it predicts the true antibody clash at AUC 0.93 (Section 6), so it earns real weight even before a global fit.

4. **Confirm the canonical DP3 metrics builder.** Determine whether `compute_metrics_mpnn.py` (currently archived) is the live RFD3+MPNN metrics producer, and promote it into `episcaf_analysis/` if so (see `docs/MIGRATION.md`).
5. **Validate the DP3 scoring path** end-to-end with `--preset antibody`, confirming `cylinder_ca_clashes` is the correct accessibility column there.
6. **Finish the 12-mer AlphaFold3 data migration** and verify landing counts against the manifest.
7. **Quantify crystal-gap dropouts.** Pull the cluster pre-filter status counts to report, per antigen, how many tiles were dropped as `crystal_epi_incomplete` (epitope residues unresolved in the deposited crystal; Section 6). This turns the recorded mechanism into a verified number, and tells us whether any intended antigen was effectively un-scorable from gaps alone.

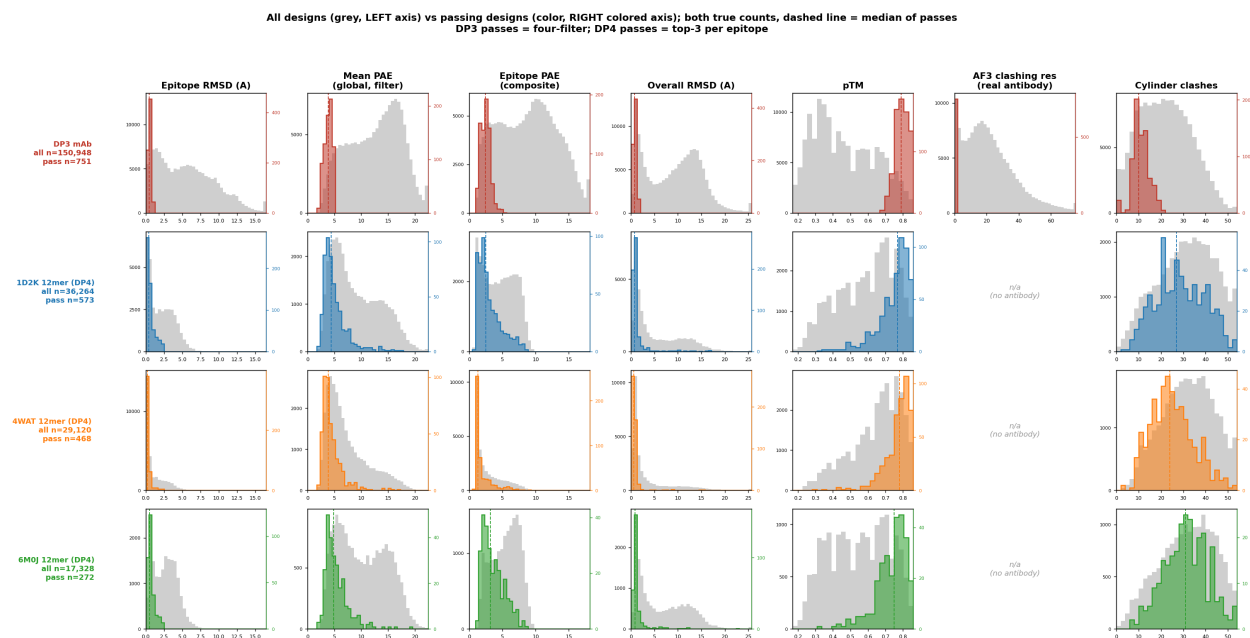


Figure 7: All designs (grey, left axis) and passing designs (color, right colored axis), both in true counts, per metric and per row; the dashed line marks the median of the passes. The twin axes keep the full population and the much smaller passing set each at its own scale, so the shape of the accepted distribution and where it lands are both visible. DP3 passes are the four-filter set; DP4 passes are the top 3 per epitope by composite score. Regenerated by `episcaf_analysis/viz/plot_passes_overlay.py`.

References